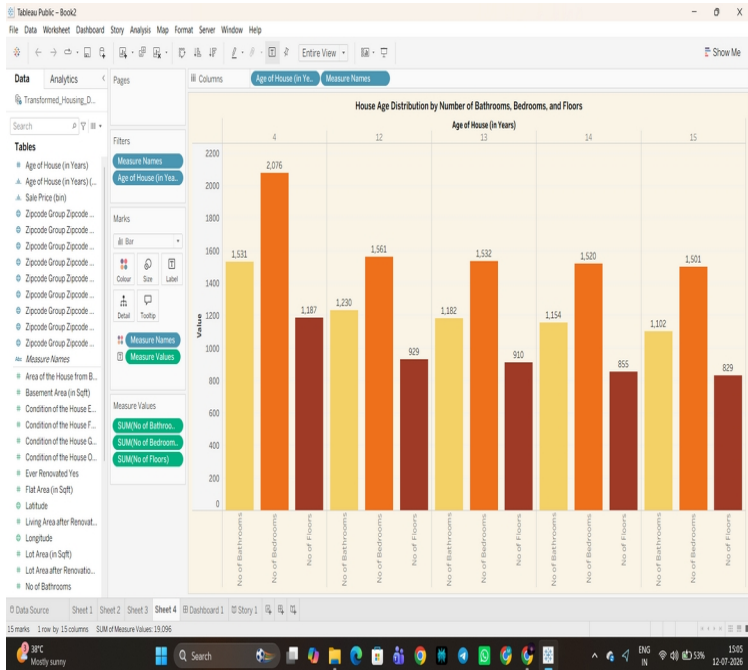
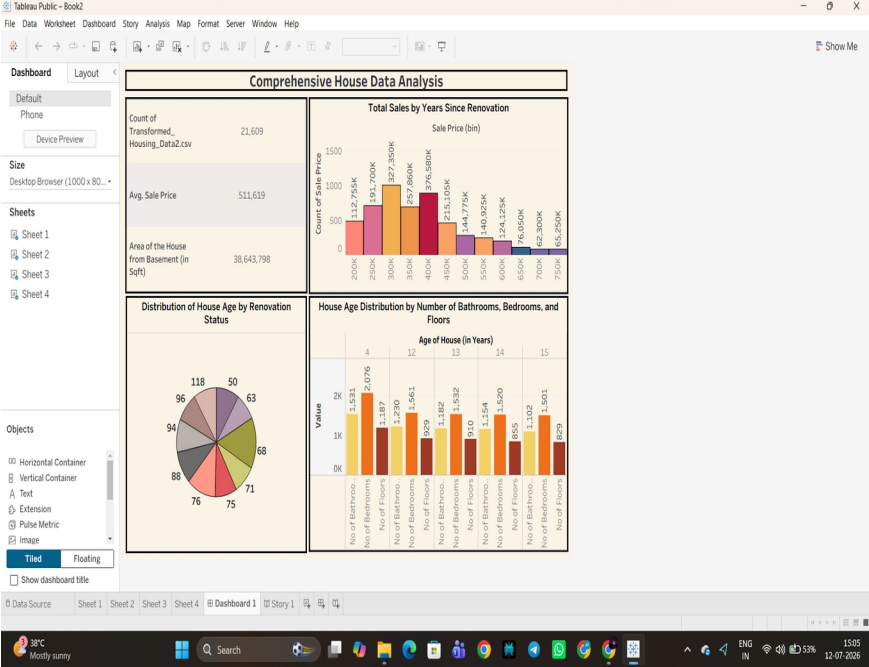


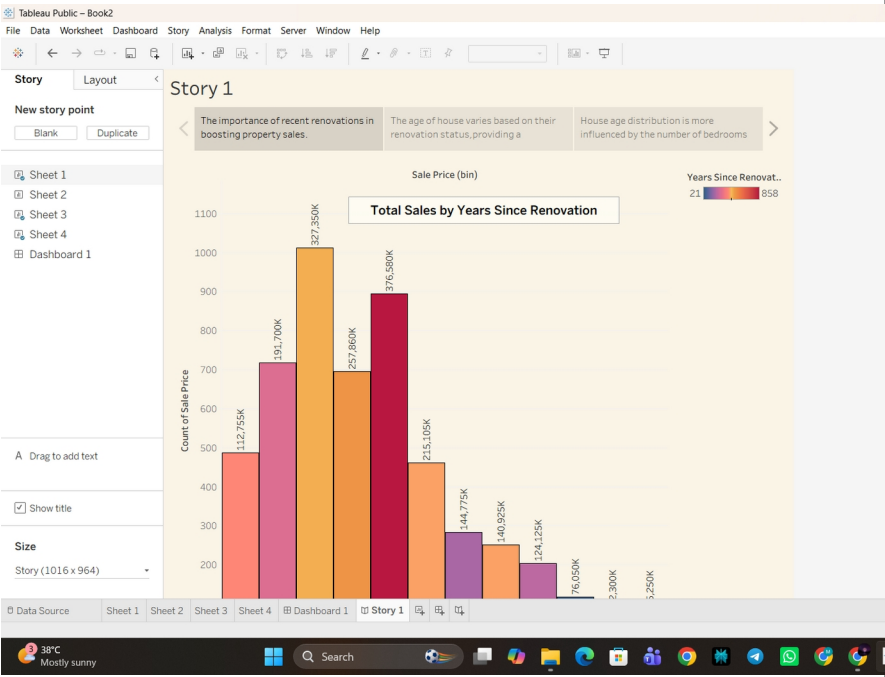
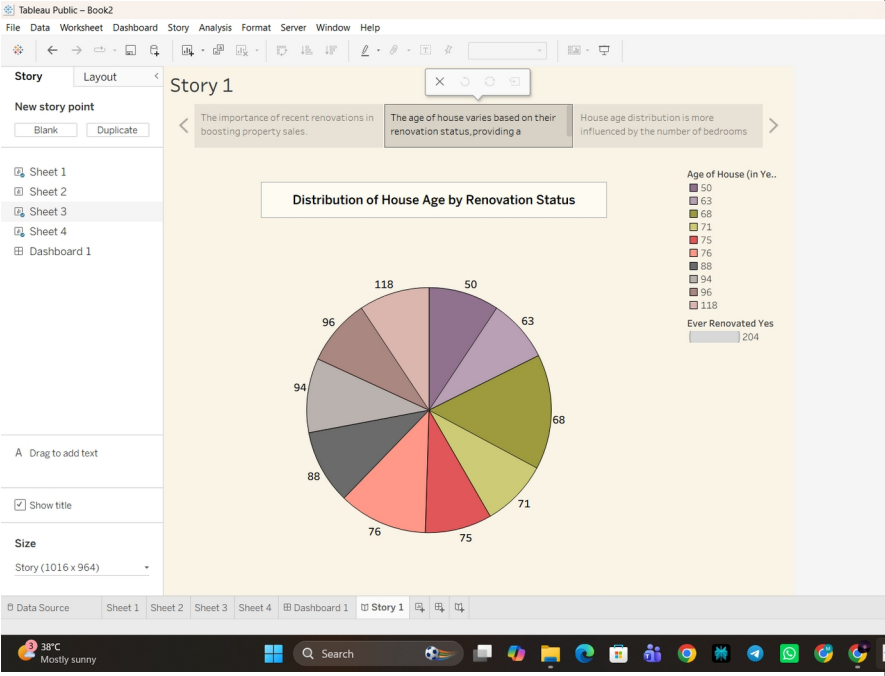
# Project Performance Testing

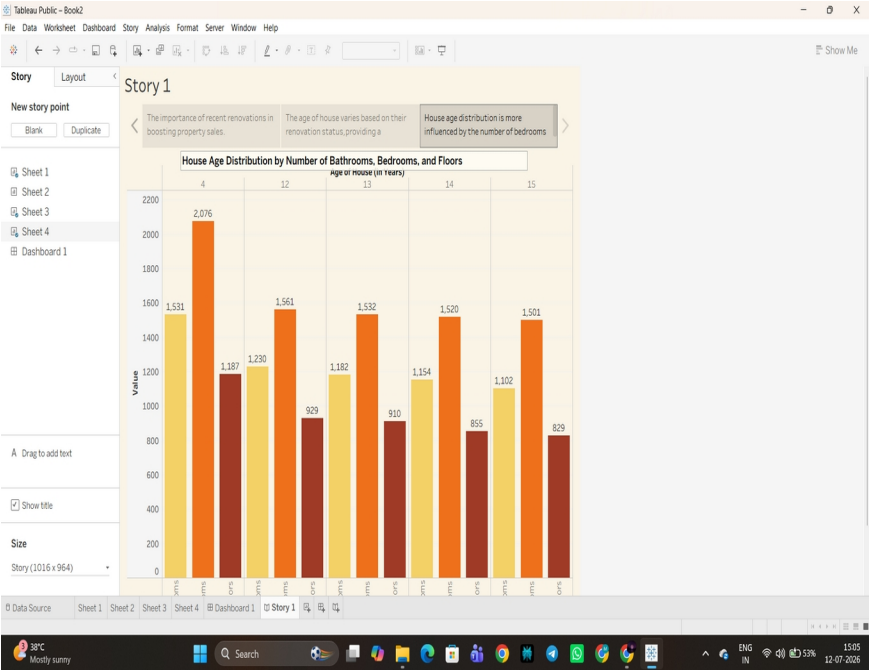
|               |                                                                                          |
|---------------|------------------------------------------------------------------------------------------|
| Date          | 13 July 2026                                                                             |
| Team ID       |                                                                                          |
| Project Name  | Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau |
| Maximum Marks | 4 Marks                                                                                  |

## Model Performance Testing:

| S.No.       | Parameter          | Screenshot / Values                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |                     |                    |                 |                     |                         |                        |                |                         |                        |                |                 |      |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |      |        |        |     |     |      |      |     |     |  |     |          |     |     |       |         |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |           |     |     |        |          |     |      |      |     |     |     |  |     |          |     |      |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |
|-------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------------------|--------------------|-----------------|---------------------|-------------------------|------------------------|----------------|-------------------------|------------------------|----------------|-----------------|------|-----|----------|-----|-----|--------|--------|-----|-----|------|-----|-----|-----|--|-----|----------|-----|------|--------|--------|-----|-----|------|------|-----|-----|--|-----|----------|-----|-----|-------|---------|-----|-----|------|-----|-----|-----|--|-----|----------|-----|-----|--------|--------|-----|-----|------|-----|-----|-----|--|-----|----------|-----|-----|--------|--------|-----|-----|------|-----|-----|-----|--|-----|-----------|-----|-----|--------|----------|-----|------|------|-----|-----|-----|--|-----|----------|-----|------|--------|--------|-----|-----|------|-----|-----|-----|--|-----|----------|-----|-----|--------|--------|-----|-----|------|-----|-----|-----|--|-----|----------|-----|-----|--------|--------|-----|-----|------|-----|-----|-----|--|------|----------|-----|-----|--------|--------|-----|-----|------|-----|-----|-----|--|------|----------|-----|-----|--------|--------|-----|-----|------|-----|-----|-----|--|------|----------|-----|-----|--------|--------|-----|-----|------|-----|-----|-----|--|
| 1           | Data Rendered      | <p>The housing dataset (Transformed_Housing_Data2.csv, ~21,609 records, 31 fields) was successfully uploaded and rendered in Tableau Public. All dataset fields, including Sale Price, Bedrooms, Bathrooms, Flat Area, Lot Area, Floors, Overall Grade, Basement Area, Age of House, Years Since Renovation, Condition of the House, Ever Renovated and Zipcode Group, were loaded correctly and made available for further analysis.</p>  <p><i>Data pane showing all housing dataset fields rendered in Tableau Public</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |                     |                    |                 |                     |                         |                        |                |                         |                        |                |                 |      |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |      |        |        |     |     |      |      |     |     |  |     |          |     |     |       |         |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |           |     |     |        |          |     |      |      |     |     |     |  |     |          |     |      |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |
| 2           | Data Preprocessing | <p><b>Removed null values</b>, verified data types, removed duplicate property records, <b>created calculated fields</b> (House Age, Years Since Renovation, Ever Renovated Yes, Sale Price bins), <b>grouped Zipcode (1–9)</b>, and <b>validated renovation and condition fields</b> before visualization.</p> <table><tr><th>Property ID</th><th>Sale_Price</th><th>No of Bedrooms</th><th>No of Bathrooms</th><th>Flat Area (in Sqft)</th><th>Lot Area (in Sqft)</th><th>No of Floors</th><th>Overall Grade</th><th>Age of House (in Years)</th><th>Years Since Renovation</th><th>Ever_Renovated</th><th>Waterfront_View</th><th>Year</th></tr><tr><td>1.0</td><td>221900.0</td><td>3.0</td><td>1.0</td><td>1180.0</td><td>5650.0</td><td>1.0</td><td>7.0</td><td>63.0</td><td>0.0</td><td>0.0</td><td>0.0</td><td></td></tr><tr><td>2.0</td><td>538000.0</td><td>3.0</td><td>2.25</td><td>2570.0</td><td>7242.0</td><td>2.0</td><td>7.0</td><td>67.0</td><td>26.0</td><td>1.0</td><td>0.0</td><td></td></tr><tr><td>3.0</td><td>180000.0</td><td>2.0</td><td>1.0</td><td>770.0</td><td>10000.0</td><td>1.0</td><td>6.0</td><td>85.0</td><td>0.0</td><td>0.0</td><td>0.0</td><td></td></tr><tr><td>4.0</td><td>604000.0</td><td>4.0</td><td>3.0</td><td>1960.0</td><td>5000.0</td><td>1.0</td><td>7.0</td><td>53.0</td><td>0.0</td><td>0.0</td><td>0.0</td><td></td></tr><tr><td>5.0</td><td>510000.0</td><td>3.0</td><td>2.0</td><td>1680.0</td><td>8080.0</td><td>1.0</td><td>8.0</td><td>31.0</td><td>0.0</td><td>0.0</td><td>0.0</td><td></td></tr><tr><td>6.0</td><td>1129575.0</td><td>4.0</td><td>4.5</td><td>5420.0</td><td>101930.0</td><td>1.0</td><td>10.0</td><td>17.0</td><td>0.0</td><td>0.0</td><td>0.0</td><td></td></tr><tr><td>7.0</td><td>257500.0</td><td>3.0</td><td>2.25</td><td>1715.0</td><td>6819.0</td><td>2.0</td><td>7.0</td><td>23.0</td><td>0.0</td><td>0.0</td><td>0.0</td><td></td></tr><tr><td>8.0</td><td>291850.0</td><td>3.0</td><td>1.5</td><td>1060.0</td><td>9711.0</td><td>1.0</td><td>7.0</td><td>55.0</td><td>0.0</td><td>0.0</td><td>0.0</td><td></td></tr><tr><td>9.0</td><td>229500.0</td><td>3.0</td><td>1.0</td><td>1780.0</td><td>7470.0</td><td>1.0</td><td>7.0</td><td>58.0</td><td>0.0</td><td>0.0</td><td>0.0</td><td></td></tr><tr><td>10.0</td><td>323000.0</td><td>3.0</td><td>2.5</td><td>1890.0</td><td>6560.0</td><td>2.0</td><td>7.0</td><td>15.0</td><td>0.0</td><td>0.0</td><td>0.0</td><td></td></tr><tr><td>11.0</td><td>662500.0</td><td>3.0</td><td>2.5</td><td>3560.0</td><td>9796.0</td><td>1.0</td><td>8.0</td><td>53.0</td><td>0.0</td><td>0.0</td><td>0.0</td><td></td></tr><tr><td>12.0</td><td>468000.0</td><td>2.0</td><td>1.0</td><td>1160.0</td><td>6000.0</td><td>1.0</td><td>7.0</td><td>76.0</td><td>0.0</td><td>0.0</td><td>0.0</td><td></td></tr></table> <p><i>Cleaned and transformed housing dataset used for dashboard visualization</i></p> | Property ID     | Sale_Price          | No of Bedrooms     | No of Bathrooms | Flat Area (in Sqft) | Lot Area (in Sqft)      | No of Floors           | Overall Grade  | Age of House (in Years) | Years Since Renovation | Ever_Renovated | Waterfront_View | Year | 1.0 | 221900.0 | 3.0 | 1.0 | 1180.0 | 5650.0 | 1.0 | 7.0 | 63.0 | 0.0 | 0.0 | 0.0 |  | 2.0 | 538000.0 | 3.0 | 2.25 | 2570.0 | 7242.0 | 2.0 | 7.0 | 67.0 | 26.0 | 1.0 | 0.0 |  | 3.0 | 180000.0 | 2.0 | 1.0 | 770.0 | 10000.0 | 1.0 | 6.0 | 85.0 | 0.0 | 0.0 | 0.0 |  | 4.0 | 604000.0 | 4.0 | 3.0 | 1960.0 | 5000.0 | 1.0 | 7.0 | 53.0 | 0.0 | 0.0 | 0.0 |  | 5.0 | 510000.0 | 3.0 | 2.0 | 1680.0 | 8080.0 | 1.0 | 8.0 | 31.0 | 0.0 | 0.0 | 0.0 |  | 6.0 | 1129575.0 | 4.0 | 4.5 | 5420.0 | 101930.0 | 1.0 | 10.0 | 17.0 | 0.0 | 0.0 | 0.0 |  | 7.0 | 257500.0 | 3.0 | 2.25 | 1715.0 | 6819.0 | 2.0 | 7.0 | 23.0 | 0.0 | 0.0 | 0.0 |  | 8.0 | 291850.0 | 3.0 | 1.5 | 1060.0 | 9711.0 | 1.0 | 7.0 | 55.0 | 0.0 | 0.0 | 0.0 |  | 9.0 | 229500.0 | 3.0 | 1.0 | 1780.0 | 7470.0 | 1.0 | 7.0 | 58.0 | 0.0 | 0.0 | 0.0 |  | 10.0 | 323000.0 | 3.0 | 2.5 | 1890.0 | 6560.0 | 2.0 | 7.0 | 15.0 | 0.0 | 0.0 | 0.0 |  | 11.0 | 662500.0 | 3.0 | 2.5 | 3560.0 | 9796.0 | 1.0 | 8.0 | 53.0 | 0.0 | 0.0 | 0.0 |  | 12.0 | 468000.0 | 2.0 | 1.0 | 1160.0 | 6000.0 | 1.0 | 7.0 | 76.0 | 0.0 | 0.0 | 0.0 |  |
| Property ID | Sale_Price         | No of Bedrooms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | No of Bathrooms | Flat Area (in Sqft) | Lot Area (in Sqft) | No of Floors    | Overall Grade       | Age of House (in Years) | Years Since Renovation | Ever_Renovated | Waterfront_View         | Year                   |                |                 |      |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |      |        |        |     |     |      |      |     |     |  |     |          |     |     |       |         |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |           |     |     |        |          |     |      |      |     |     |     |  |     |          |     |      |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |
| 1.0         | 221900.0           | 3.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1.0             | 1180.0              | 5650.0             | 1.0             | 7.0                 | 63.0                    | 0.0                    | 0.0            | 0.0                     |                        |                |                 |      |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |      |        |        |     |     |      |      |     |     |  |     |          |     |     |       |         |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |           |     |     |        |          |     |      |      |     |     |     |  |     |          |     |      |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |
| 2.0         | 538000.0           | 3.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2.25            | 2570.0              | 7242.0             | 2.0             | 7.0                 | 67.0                    | 26.0                   | 1.0            | 0.0                     |                        |                |                 |      |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |      |        |        |     |     |      |      |     |     |  |     |          |     |     |       |         |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |           |     |     |        |          |     |      |      |     |     |     |  |     |          |     |      |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |
| 3.0         | 180000.0           | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1.0             | 770.0               | 10000.0            | 1.0             | 6.0                 | 85.0                    | 0.0                    | 0.0            | 0.0                     |                        |                |                 |      |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |      |        |        |     |     |      |      |     |     |  |     |          |     |     |       |         |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |           |     |     |        |          |     |      |      |     |     |     |  |     |          |     |      |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |
| 4.0         | 604000.0           | 4.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3.0             | 1960.0              | 5000.0             | 1.0             | 7.0                 | 53.0                    | 0.0                    | 0.0            | 0.0                     |                        |                |                 |      |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |      |        |        |     |     |      |      |     |     |  |     |          |     |     |       |         |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |           |     |     |        |          |     |      |      |     |     |     |  |     |          |     |      |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |
| 5.0         | 510000.0           | 3.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2.0             | 1680.0              | 8080.0             | 1.0             | 8.0                 | 31.0                    | 0.0                    | 0.0            | 0.0                     |                        |                |                 |      |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |      |        |        |     |     |      |      |     |     |  |     |          |     |     |       |         |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |           |     |     |        |          |     |      |      |     |     |     |  |     |          |     |      |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |
| 6.0         | 1129575.0          | 4.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 4.5             | 5420.0              | 101930.0           | 1.0             | 10.0                | 17.0                    | 0.0                    | 0.0            | 0.0                     |                        |                |                 |      |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |      |        |        |     |     |      |      |     |     |  |     |          |     |     |       |         |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |           |     |     |        |          |     |      |      |     |     |     |  |     |          |     |      |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |
| 7.0         | 257500.0           | 3.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2.25            | 1715.0              | 6819.0             | 2.0             | 7.0                 | 23.0                    | 0.0                    | 0.0            | 0.0                     |                        |                |                 |      |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |      |        |        |     |     |      |      |     |     |  |     |          |     |     |       |         |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |           |     |     |        |          |     |      |      |     |     |     |  |     |          |     |      |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |
| 8.0         | 291850.0           | 3.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1.5             | 1060.0              | 9711.0             | 1.0             | 7.0                 | 55.0                    | 0.0                    | 0.0            | 0.0                     |                        |                |                 |      |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |      |        |        |     |     |      |      |     |     |  |     |          |     |     |       |         |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |           |     |     |        |          |     |      |      |     |     |     |  |     |          |     |      |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |
| 9.0         | 229500.0           | 3.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1.0             | 1780.0              | 7470.0             | 1.0             | 7.0                 | 58.0                    | 0.0                    | 0.0            | 0.0                     |                        |                |                 |      |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |      |        |        |     |     |      |      |     |     |  |     |          |     |     |       |         |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |           |     |     |        |          |     |      |      |     |     |     |  |     |          |     |      |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |
| 10.0        | 323000.0           | 3.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2.5             | 1890.0              | 6560.0             | 2.0             | 7.0                 | 15.0                    | 0.0                    | 0.0            | 0.0                     |                        |                |                 |      |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |      |        |        |     |     |      |      |     |     |  |     |          |     |     |       |         |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |           |     |     |        |          |     |      |      |     |     |     |  |     |          |     |      |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |
| 11.0        | 662500.0           | 3.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2.5             | 3560.0              | 9796.0             | 1.0             | 8.0                 | 53.0                    | 0.0                    | 0.0            | 0.0                     |                        |                |                 |      |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |      |        |        |     |     |      |      |     |     |  |     |          |     |     |       |         |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |           |     |     |        |          |     |      |      |     |     |     |  |     |          |     |      |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |
| 12.0        | 468000.0           | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1.0             | 1160.0              | 6000.0             | 1.0             | 7.0                 | 76.0                    | 0.0                    | 0.0            | 0.0                     |                        |                |                 |      |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |      |        |        |     |     |      |      |     |     |  |     |          |     |     |       |         |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |           |     |     |        |          |     |      |      |     |     |     |  |     |          |     |      |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |     |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |      |          |     |     |        |        |     |     |      |     |     |     |  |

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| 3 | <b>Utilization of Filters</b>  | <p>Filters used: Zipcode Group (1–9), Age of House (in Years), Sale Price (bin), No of Bedrooms, No of Bathrooms and Measure Names.</p> <p>Aggregation functions: SUM(Sale Price) and CNT(Sale Price) were primarily used for the KPI totals (Total Properties, Total Basement Area), while AVG(Sale Price) was applied for the Avg. Sale Price KPI card. SUM(Ever Renovated Yes) drives the renovation-status pie chart, and SUM(No of Bedrooms), SUM(No of Bathrooms) and SUM(No of Floors) drive the house-age distribution chart.</p> <p>Worksheets utilize coordinated filters — the Age of House (in Years) filter and Sale Price (bin) filter are applied consistently across Sheet 1, Sheet 3 and Sheet 4 — to allow synchronized cross-filtering between the KPI cards, the renovation-impact chart and the age-distribution chart on the dashboard.</p>                                                                                                                                                                                         |
| 4 | <b>Calculation fields Used</b> | <p>Calculated fields created and used in this visualization include House Age, Years Since Renovation, Ever Renovated Yes (renovation-status flag), Sale Price (bin) and Zipcode Group.</p> <p><b>Logic / Syntax:</b></p> <ul style="list-style-type: none"><li>House Age = DATEDIFF('year', [Year Built], TODAY())</li><li>Years Since Renovation = IF [Year Renovated] = 0 THEN [House Age] ELSE DATEDIFF('year',[Year Renovated],TODAY()) END</li><li>Ever Renovated Yes = IF [Year Renovated] &gt; 0 THEN 'Yes' ELSE 'No' END</li><li>Sale Price (bin) = created using Tableau's Create Bins on Sale Price, with a fixed bin size for the sales-distribution chart</li><li>Zipcode Group = raw zipcodes bucketed into 9 calculated groups (Zipcode Group 1–9) for location-wise analysis</li></ul> <p>Purpose: These calculated fields let the KPI cards, the Total Sales by Years Since Renovation chart, and the renovation/age breakdowns aggregate correctly at the dashboard level, without requiring row-level detail from the raw dataset.</p> |
| 5 | <b>Dashboard design</b>        | <p>All visualizations in use — Dashboard Design (Comprehensive House Data Analysis):</p>  <p><i>Comprehensive House Data Analysis — Tableau Public Dashboard</i></p> <p>The dashboard combines the KPI summary panel (Total Properties, Avg. Sale Price, Area of the House from Basement) with the Total Sales by Years Since Renovation chart, the Distribution of House Age by Renovation Status pie chart, and the House Age Distribution by Number of Bathrooms, Bedrooms and Floors</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

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|---|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                     | chart.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6 | <b>Story Design</b> | <p>Visualizing Housing Market Trends using a Tableau Story. Multiple story points were created to narrate the housing-market findings in sequence: the importance of recent renovations in boosting property sales, how house age varies by renovation status, and how house-age distribution is influenced by the number of bedrooms.</p>  <p><b>Story Point 1 — House Age Distribution by Number of Bathrooms, Bedrooms and Floors</b></p>  <p><b>Story Point 2 — Distribution of House Age by Renovation Status</b></p> |



Story Point 3 — Total Sales by Years Since Renovation